



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

UNIX PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Operating Systems

TOTAL: 10 QUESTIONS

Q1 What is Unix and what problem in Operating Systems does it solve? **Does it**

Q6 How does Unix handle reliability and high availability? **Observability**

Q2 What are the core components or concepts in Unix? **Architecture**

Q7 What observability does Unix provide out of the box? **Lifecycle**

Q3 How does Unix compare to its main alternatives? **Configuration**

Q8 What are common performance bottlenecks in Unix? **Compliance**

Q4 How do you get started with Unix for a new project? **Hardening**

Q9 What security best practices apply when running Unix? **Testing**

Q5 What configuration options most affect Unix's behavior in production? **ISO / IdP**

Q10 What mistakes do teams commonly make when adopting Unix? **Tuning**



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Unix is a tool or platform that

Mitigation

Unix is a tool or platform that automates or simplifies a specific concern in the Operating Systems domain, reducing manual effort and operational complexity for engineering teams.

A6 Unix provides HA through leader election, replication, and observability

Unix provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 Unix has key building blocks that work

Primitives

Unix has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Unix exposes metrics (Prometheus endpoint or built-in dashboard) and automation

Unix exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 Unix trades off simplicity against feature richness

Hardening

Unix trades off simplicity against feature richness differently than its alternatives. Choose Unix when its specific strengths—performance, ecosystem, or ease of use—align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

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A4 Install Unix via its package manager or

Gotchas

Install Unix via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run Unix with the principle of least

Assessment

Run Unix with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

Concurrency

A10 Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.

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